MODERN APPLICATION DEVELOPMENT 1

PROJECT REPORT

1. Student Details

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About Me

I am a student in the IIT Madras BS Degree Program with a strong interest in web application development and backend systems. I have experience working with technologies such as Flask, databases, and responsive UI design. Through this project, I gained practical understanding of database schema design, authentication systems, and role-based web applications used in modern software development.

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2. Project Details

Project Title

CareerBridge – Placement Portal Application

Problem Statement

Institutes require an efficient system to manage campus recruitment activities involving companies, students, and placement drives. Many institutes rely on spreadsheets, emails, or manual processes, which makes it difficult to manage company approvals, student applications, and placement records.

The objective of this project is to develop a Placement Portal Application that allows Admin, Companies, and Students to interact through a centralized recruitment platform.

Approach

The application was developed using Flask as the backend framework and SQLite as the database. Role-based authentication ensures different functionalities for Admin, Company, and Student users.

Development steps included:

- Designing database schema using SQLAlchemy ORM
- Implementing dashboards for Admin, Company, and Student
- Placement drive creation and approval workflow
- Student job application and status tracking
- Responsive UI using Bootstrap

The system follows a modular Flask architecture separating backend logic, templates, and static resources.

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3. AI / LLM Declaration

AI tools such as ChatGPT were used for limited assistance during development for:

- Debugging errors
- Improving code structure
- Documentation formatting
- Understanding Flask implementation

Approximate AI usage: 20–25%.

All core components including database design, backend logic, and system integration were implemented manually.

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4. Technologies Used

Technology	Purpose
Python	Programming language
Flask	Backend framework
SQLite	Database system
SQLAlchemy	ORM
Jinja2	Template engine
HTML	Page structure
CSS	Styling
Bootstrap	Responsive UI
Flask-Login	Authentication

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5. Database Schema

The application uses SQLite database with SQLAlchemy ORM.

User: id, name, email, password, role, is_active, is_approved
StudentProfile: id, roll_no, course, cgpa, resume_link, graduation_year, skills, phone, address, github, linkedin, user_id

CompanyProfile: id, company_name, website, description, industry, user_id
JobPosition: id, title, description, skills, experience, salary, location, is_active, is_approved, company_id
Application: id, student_id, job_id, status, applied_at, updated_at
Placement: id, application_id, joining_date, package
Notification: id, student_id, job_id, title, message, is_read, created_at

Relationships

- **User** → StudentProfile (1:1)
- **User** → CompanyProfile (1:1)
- **CompanyProfile** → JobPosition (1:N)
- **StudentProfile** → Application (1:N)
- **JobPosition** → Application (1:N)
- **Application** → Placement (1:1)

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6. API Endpoints

Endpoint	Method	Description
/login	POST	Authenticate user
/register/student	POST	Register student
/register/company	POST	Register company
/admin/dashboard	GET	Admin dashboard
/jobs	GET	View placement drives
/apply/<job_id>	POST	Apply for job
/applications	GET	View applications
/update_status	POST	Update status

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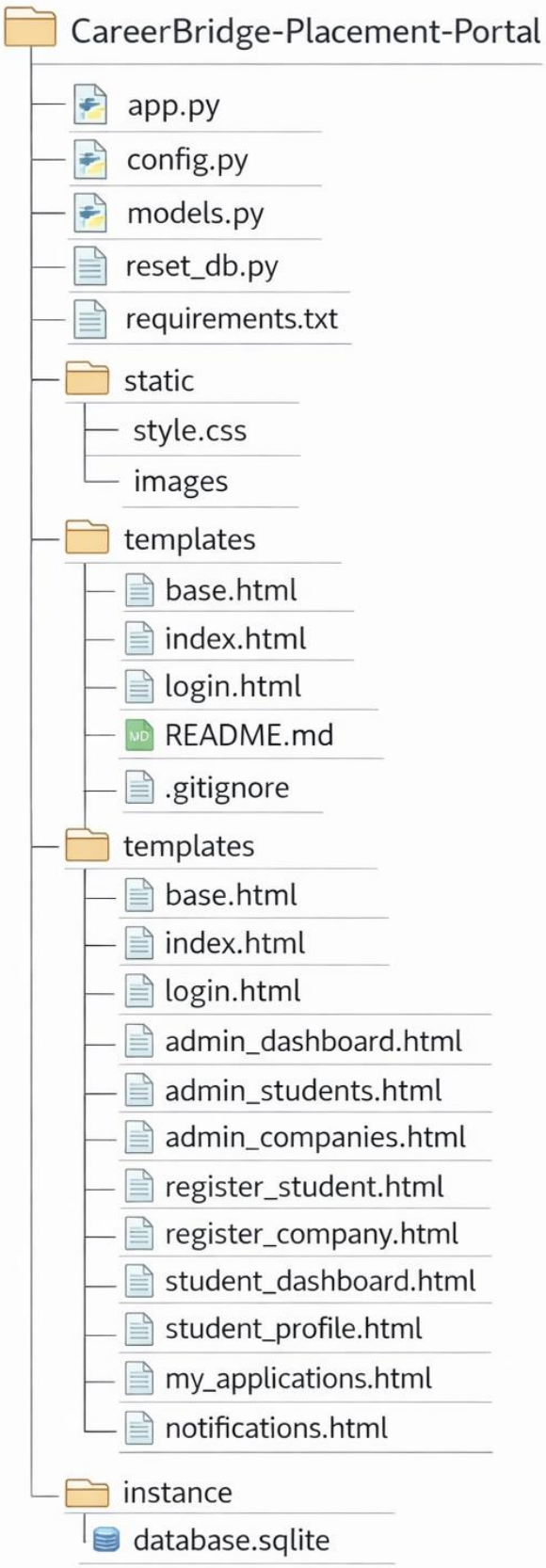
7. Architecture & Features

Folder Structure



Main Features

Admin

- Approve companies and placement drives
- View students, companies, applications
- Search and deactivate accounts

Company

- Create placement drives
- View applicants
- Shortlist or reject candidates

Student

- Register and update profile
- Upload resume
- Browse placement drives
- Apply for jobs and track status

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8. Video Presentation

Drive Link:
(Add your Google Drive video link here)

The video demonstrates:

- Project overview
- System architecture
- Admin, Company, and Student dashboards
- Job application workflow

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9. Conclusion

The **CareerBridge** Placement Portal provides a centralized platform for managing campus recruitment activities efficiently. The system improves placement management, student applications, and company interactions.This project demonstrates practical implementation of Flask web development, relational database design, and role-based authentication.